

Teaching Monocular Depth the Metre with the Phone’s Own LiDAR: Fine-Tuned vs. Pretrained Depth Anything V2 for Room Reconstruction, Measured Against a Laser Scanner

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Abstract—Most phones have no LiDAR and must scan rooms with monocular depth, yet off-the-shelf “metric” models such as Depth Anything V2 (DA-v2) Metric-Indoor [1], trained on synthetic images, misplace a real room by 54.3 cm. We ask whether fine-tuning such a model with the phone’s own LiDAR depth, with no laser scanner involved, makes it more accurate than its pretrained self, measured against the Faro Focus S70 laser scanner [2] that provides the ground truth of ARKitScenes [3]. We fine-tune only the DPT head [4] (the depth decoder; the image encoder stays frozen) on 24 rooms with iPad LiDAR labels, giving FT-LiDAR-24, and compare it with the pretrained model in one pipeline—per-frame depth fused into a truncated signed distance function (TSDF) volume [5], then meshed—where the depth source is the only variable, on rooms never used for training: six with a laser-scanned reference mesh and 20 more evaluated per frame. FT-LiDAR-24 lowers the Chamfer distance [6] (mean nearest-point distance between the reconstructed and reference meshes, both ways) from 54.3 cm to 15.7 cm ($3.5\times$, better in every room) and AbsRel [7] (mean relative depth error) from 0.410 to 0.130 (better in 19/20 rooms), with no calibration or sensor at run time; a laser teacher is not measurably better. Its remaining error is a scale that swings from view to view, which fine-tuning does not remove, so it still trails iPad LiDAR (3.0 cm) and falls short of overview accuracy ($\geq 90\%$ of the room within 10 cm).

Index Terms—monocular depth estimation, transfer learning, metric depth, LiDAR supervision, TSDF fusion, 3D room reconstruction, ARKitScenes

I. INTRODUCTION

Scanning a room into a 3D model is already a consumer feature on iPhone Pro and iPad Pro (e.g., Apple RoomPlan [8]), and it relies on the *LiDAR sensor* for depth in metres. Devices with LiDAR are, however, a minority: non-Pro iPhones and almost every Android phone have none. Their alternative is *monocular depth estimation*—depth predicted from a single colour image—which has advanced rapidly from MiDaS [9] to Depth Anything [1], [10]. A product team must therefore know **by how many centimetres the room is wrong if LiDAR is replaced by a monocular model, and which use cases that still serves**. Standard per-image benchmarks [7], which average the depth error over the pixels of single images, do not answer this, and end-to-end reconstruction systems [11]–

[13] cannot, because they design the whole pipeline around their own model.

Some recent models output metres directly [1], [14], [15], but they are trained on images that do not come from a phone camera—DA-v2 Metric-Indoor on rendered rooms of the Hypersim dataset [16]—and in our experiments a room reconstructed with DA-v2 Metric-Indoor is 54.3 cm wrong on average. The direct remedy is to *fine-tune* the model on depth from the target device, which raises the question of where these labels come from: a survey-grade laser scanner [2], which few app makers own, or the *LiDAR already inside Pro devices*, whose depth an app can collect during normal use. If the latter works, devices with LiDAR can teach those without it.

We therefore ask whether DA-v2 Metric-Indoor fine-tuned with iPad LiDAR depth, with no laser scanner at training time, is more accurate than the same model pretrained, when both are measured against the Faro laser scanner that provides the ground truth of ARKitScenes [3] on rooms never used for training. The study has four objectives: (1) to fine-tune the model on iPad LiDAR depth from 24 rooms, giving FT-LiDAR-24, and compare it with the pretrained model on held-out rooms, counting it as better if its mean error is lower and the per-room difference is significant at the 5% level under an exact Wilcoxon signed-rank test [17]; (2) to measure whether the source of the labels matters, by comparing Faro with LiDAR labels on the same 10 rooms (FT-Faro-10 and FT-LiDAR-10), 10 with 24 rooms of LiDAR labels, and a model that uses the camera’s focal length instead of in-domain labels (Depth Pro [15]); (3) to place FT-LiDAR-24 among the depth sources available at run time, namely iPad LiDAR and a relative model scaled per frame on sparse points or once per room; and (4) to find the error that fine-tuning leaves, by fitting a depth scale to every frame and testing whether it follows time, which would indicate drift, or what the view shows.

Every comparison is scored in two ways. As a room mesh, it is scored by the Chamfer distance [6], the mean distance

from each surface to the nearest point of the other, and by the F-score at 5 cm (F@5 cm) [18], which combines the share of the reconstruction lying within 5 cm of the reference with the share of the reference reconstructed within 5 cm. Per frame, it is scored by AbsRel [7], the mean relative depth error, and by the share of pixels whose predicted depth is within a factor of 1.25 of the true depth, which we call within-1.25× [7]. For objective (3), the share of the reference reconstructed within 2, 5 and 10 cm stands for the accuracy needed for renovation measurement, furniture placement and a room overview, respectively, and a use case counts as served when that share reaches 90 %.

The scope is as follows. The data are home rooms of ARKitScenes captured with one iPad Pro model (colour images of 640×480 pixels, LiDAR depth of 256×192); the ground truth is the Faro depth distributed with the dataset, and iPad LiDAR is never used as ground truth. Only the DPT head [4], the depth decoder of DA-v2 Metric-Indoor Large, is fine-tuned, with one training recipe, while its image encoder stays frozen. Six test rooms are evaluated as meshes and per frame, and 20 further rooms per frame only, all disjoint from the rooms used for training and checkpoint selection. Every depth source passes through the same offline pipeline—per-frame depth, scale alignment, fusion into a truncated signed distance function (TSDF) volume [5], meshing—so the depth source is the only variable, and because every source and the reference share the camera poses recorded by ARKit, pose error is excluded. Learned reconstruction systems [11]–[13], other phone cameras including Android, real-time inference on the device and pose estimation are outside the scope.

Each objective has a matching expected benefit. The first tells an app maker whether data it can already collect from Pro-device users improves room scans on devices without LiDAR, with no calibration or extra sensor at run time; the second, whether labelling needs a laser scanner at all; the third, which product features a device without LiDAR can offer, together with a pipeline in which a new depth source is evaluated by changing one configuration entry; and the fourth, where to invest next—per-frame scale, training data or model size.

II. RELATED WORK

Relative monocular depth. MiDaS [9] trains across datasets with a scale- and shift-invariant loss, so its output is disparity up to an affine transform: two constants (s, t) per image are needed for metres. Depth Anything [10] and V2 [1] scale this idea with large unlabelled corpora. We treat (s, t) as an experimental variable (the *ScaleAligner*) rather than hiding it in the model.

Metric monocular depth. ZoeDepth [14] and the metric DA-v2 variants (fine-tuned on Hypersim / Virtual KITTI) output metres directly. Metric3D [19], UniDepth [20] and Depth Pro [15] instead condition on, or estimate, camera intrinsics. These works report per-image metrics on NYUv2/KITTI; we show that on real room video the per-frame scale of a metric model still swings widely within one scan—invisible to per-image metrics, obvious once frames are fused. The closest

TABLE I
SCALE ALIGNERS. EVERY NON-IDENTITY ALIGNER FITS (s, t) IN INVERSE DEPTH WITH TRIMMED LEAST SQUARES.

Aligner	Fitted on / meaning
identity	nothing; for Faro, LiDAR and metric models
oracle_frame	that frame’s GT; the model’s ceiling (not deployable)
per_scene	GT of 10 frames spread over the scan, fitted once; upper bound of a one-off calibration
sparse_points	200 LiDAR pixels per frame at confidence 2, a proxy for VIO feature points; upper bound of a VIO-based system

work in data is Prompt Depth Anything [21], which feeds iPhone LiDAR to the network *at run time* on ARKitScenes; we instead use LiDAR only as a *training* teacher, so the deployed model needs no depth sensor.

Depth fusion and learned reconstruction. TSDF fusion [5], [22] neither knows nor cares where metric, posed depth comes from, making it a fair fixed stage for every source; we use Open3D [23]. Atlas [11], NeuralRecon [12] and SimpleRecon [13] learn depth or TSDF from multiple images and are strong ScanNet baselines, but do not answer what room a per-image model plus the device’s VIO pose yields, so we do not compare against them. Scale recovery from sparse tracked points follows CNN-SLAM [24] and MonoSDF [25]; our *sparse_points* aligner belongs to this line.

Mobile LiDAR and products. ARKitScenes [3] is the first dataset releasing both iPad LiDAR depth and Faro laser depth for the same rooms. RoomPlan [8] and similar apps require LiDAR, and no public report states how many centimetres they deviate from a laser scanner.

III. SYSTEM DESIGN

The research question (“change only the depth source—how much changes?”) leads to one design rule: every depth source passes through the same code after the point where it is produced, and whatever can be swapped is swapped from a YAML config, so one result table is one config file (Fig. 1).

DepthSource. An interface `get_depth(frame)` returning an (H, W) depth map and a flag `is_metric`. Implementations read Faro depth (`gt`), ARKit LiDAR (`lidar`) or run a monocular network (`mono`: DA-v2, MiDaS, fine-tuned checkpoints) on the upright RGB frame.

ScaleAligner. A relative model is correct only up to an affine map, and *how* that map is obtained is a system decision measured separately from the model (Table I). Affine-invariant models are affine in *disparity*, so every aligner fits $1/gt \approx s \cdot (1/pred) + t$ by trimmed least squares and inverts back; on the development scene this change alone moved `mono+oracle` from 15.2 cm to 5.4 cm. Metric sources must use `identity`; the pipeline refuses to build otherwise.

Fixed stages. Open3D `ScalableTSDFVolume` with 4 cm voxels, truncation $3 \times$ voxel and depth range 0.1–5 m; every source is resampled to the LiDAR resolution (256×192) for

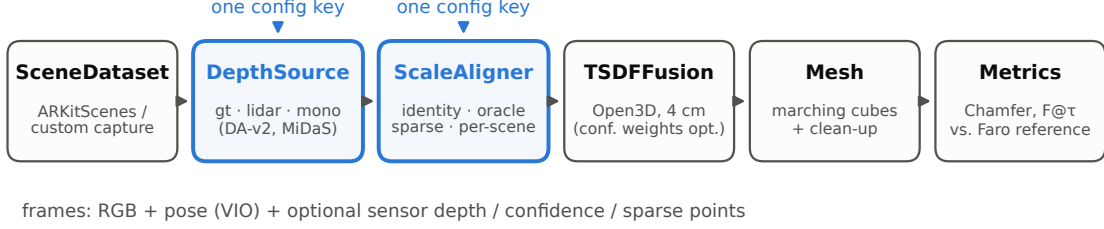


Fig. 1. Pipeline. The two highlighted stages are the only things that change between rows of a result table; everything to their right is fixed.

TABLE II
TEST SCENES (3D AND 2D). FRAMES = FARO FRAMES EVERY ROW IS EVALUATED ON.

video_id	Room	Extent (m)	Frames
42444474	kitchen/dining, glass (dev. scene)	$9.9 \times 5.1 \times 4.2$	236
47333774	standard bedroom	$5.1 \times 4.9 \times 3.3$	375
47115299	large living room	$8.1 \times 6.1 \times 2.5$	324
42897743	bathroom (mirror, tiles)	$8.0 \times 3.4 \times 3.6$	448
47429736	bedroom + closets	$6.6 \times 6.4 \times 2.7$	504
45261556	kitchen (glossy cabinets)	$7.1 \times 6.1 \times 2.4$	465

TABLE III
EXPERIMENTS. EXP. 6–7 ANSWER THE MAIN QUESTION; EXP. 1–5 GIVE CONTEXT.

Exp.	Variable (everything else fixed)
6	pretrained vs. FT-LiDAR-24 ; ablation FT-Faro-10 / FT-LiDAR-10 / Depth Pro (6 scenes, 3D + 2D)
7	pretrained / FT-LiDAR-24 / LiDAR on 20 further rooms (2D)
1	depth source $\times$ aligner: gt / lidar / DA-v2 L + {oracle, sparse, per_scene} / metric
2–5	voxel size; frame stride; model size; LiDAR confidence

fairness, and only pixels where Faro has depth are integrated, so no source is penalised for seeing beyond the reference. Marching cubes and small-cluster removal follow. Poses are the dataset’s ARKit VIO trajectory for every row.

IV. EXPERIMENTAL SETUP

A. Dataset and reference

ARKitScenes [3] is the only dataset that provides, for the same room, RGB video from a real device (iPad Pro, 640×480), that device’s LiDAR depth (256×192) and Faro laser depth registered to the camera poses (1920×1440 , ~ 4 fps). **Ground truth is the Faro data as distributed.** In 2D we compare per-frame depth with Faro depth directly (nearest-neighbour down-sampling, so every GT value is a measured one). In 3D, since no laser mesh is shipped, we fuse Faro depth from every frame at 1 cm voxels once per scene into a reference mesh. The `gt` row at 4 cm therefore has non-zero error—the pipeline’s own discretisation loss—and because the reference uses the same poses as every row, pose drift is excluded and all differences are due to the depth source. iPad LiDAR is never used as ground truth.

The six test scenes (Table II) span standard, large and glossy/mirror rooms. Scene 42444474 was used during development, so its numbers are not held-out; per-scene results show the conclusions hold on the other five.

B. Fine-tuning protocol

Scenes disjoint from the test set at the `visit_id` level were drawn with seed 0: 10 training scenes with both Faro and LiDAR, 14 more with LiDAR only, and 3 Validation-fold scenes for checkpoint selection. Three fine-tunes share one recipe: **FT-Faro-10** uses Faro labels on the 10 scenes

(3594 frames), **FT-LiDAR-10** iPad LiDAR (confidence ≥ 1) on the same 10 scenes (8656 frames), and **FT-LiDAR-24, the main model**, LiDAR on all 24 scenes (19 896 frames). FT-Faro-10 and FT-LiDAR-10 do not share frames: Faro exists at ~ 4 fps, LiDAR with every colour frame (we take every third, ~ 10 fps), so FT-LiDAR-10 sees $\sim 2.4\times$ more distinct views than FT-Faro-10. The DINOv2 encoder is frozen; only the DPT neck and metric head train, with SiLog loss, AdamW ($1r\ 5 \times 10^{-5}$, cosine schedule), batch 8 (2×4 accumulation) on 518×518 crops, for 6000 steps (FT-Faro-10, FT-LiDAR-10) or 12 000 (FT-LiDAR-24), keeping the checkpoint with the lowest validation AbsRel. A 6000-step run takes about an hour on an RTX 3070. At test time the fine-tuned models are plain DepthSource rows: RGB in, metres out, identity aligner, same input preparation as the pretrained model.

C. Metrics and experiments

3D: 200k points sampled on both meshes; accuracy = mean $\text{pred} \rightarrow \text{ref}$ distance, completeness = $\text{ref} \rightarrow \text{pred}$, Chamfer = their mean; precision/recall/F-score at $\tau = 2/5/10$ cm, with F@5 cm as headline (5 cm $\approx$ furniture-placement tolerance). **2D**: per-frame AbsRel, within- $1.25\times$ and RMSE against Faro depth; pixels are selected by the ground truth (0.1–5 m) and predictions clipped to that range, averaged per room then over rooms. Table III lists the experiments; numbers are mean $\pm$ std over the six scenes. **Exp. 7** adds 20 Validation-fold rooms (seed 0, one video per visit, excluding every visit of the training, validation and test sets) evaluated in 2D only on every second Faro frame; these rooms decided nothing.

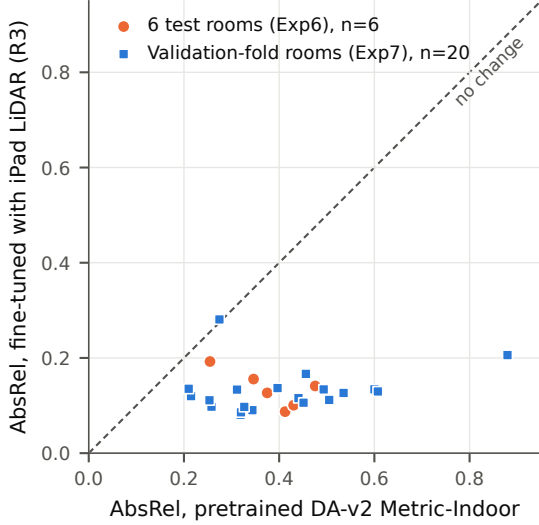


Fig. 2. Per-room AbsRel against Faro: pretrained (x) vs. FT-LiDAR-24 (y), six test rooms and 20 Validation-fold rooms. Below the diagonal = the fine-tune is better.

V. RESULTS

All numbers are generated from the per-run `metrics.json` files; units are cm and F is $F@5$ cm.

A. Main result: FT-LiDAR-24 vs. the pretrained model

The two rows differ only in the weights of the DPT neck and metric head; at test time both see the same frames and input with no alignment of any kind, and no room here was used for training or checkpoint selection.

- 1) **FT-LiDAR-24 beats the pretrained model in every test room.** Chamfer 54.3 cm $\rightarrow$ 15.7 cm ($3.5\times$; 38.6 ± 8.6 cm better per room, 6/6 rooms, exact Wilcoxon $p = 0.031$, the smallest attainable at $n = 6$) and $F@5$ cm $0.03 \rightarrow 0.24$ (Tables IV, V). Accuracy falls more than completeness: the pretrained model builds the room too far away (15–31 % over-estimation, Section VI-B); FT-LiDAR-24 brings surfaces back near where they belong.
- 2) **The same holds on rooms that took part in nothing.** On the 20 Validation-fold rooms AbsRel drops $0.410 \rightarrow 0.130$ ($3.2\times$; better in 19/20 rooms, exact Wilcoxon $p = 3.8 \times 10^{-6}$) and the within- $1.25\times$ share rises $0.333 \rightarrow 0.827$ (Table VI, Fig. 2). The one exception, 48018730 (AbsRel $0.275 \rightarrow 0.281$), is one of the pretrained model’s best rooms; in the other 19, FT-LiDAR-24’s AbsRel is 21–64 % of the pretrained one.
- 3) **It does not reach a sensor.** FT-LiDAR-24 is ~ 13 cm behind iPad LiDAR in Chamfer and 9.7 cm behind a per-frame fit on 200 sparse points; fine-tuning fixes the scale of the *domain*, not of the *frame being looked at* (Section VI-A).



Fig. 3. Exp. 6 top-down error maps (rows = scenes; columns = Depth Pro, the three fine-tunes and the pretrained metric model, as labelled above each panel; colour = distance to the reference, 0–10 cm). The pretrained column is red almost everywhere—the room is built at the wrong distance; after fine-tuning the error concentrates at object edges and far walls.

B. Teacher ablation and a no-fine-tuning baseline

The laser teacher is not measurably better than the LiDAR teacher. FT-Faro-10 14.6 cm, FT-LiDAR-24 15.7 cm and FT-LiDAR-10 16.6 cm all lie within the across-room std (4.0–4.7 cm), and no row wins every room (FT-Faro-10 beats FT-LiDAR-24 in three rooms, FT-LiDAR-24 wins two, one tie; Table V, Fig. 3). Because FT-Faro-10 and FT-LiDAR-10 also differ in the number of views, the honest claim is that a phone’s depth sensor can replace the laser as the teacher at no measurable cost on these rooms—not that the two are equal. Fourteen extra LiDAR-only rooms (FT-LiDAR-24 vs. FT-LiDAR-10) do not help measurably either; we keep FT-LiDAR-24 as the main model because it was best on validation and is the only recipe that scales without a laser.

Knowing the camera does not rescue metric depth. Depth Pro [15], given the true focal length and no fine-tuning, reaches 155.4 cm ($F@5$ cm 0.00), worse than the pretrained DA-v2 Metric row: it predicts a 48° field of view where the camera has 62° , and even after removing all scale per frame its AbsRel is 0.24–0.54. We verified the wrapper on a synthetic room with exact ground truth; the claim is about this model on this low-resolution indoor input, not about intrinsics-conditioned models in general.

TABLE IV

MAIN RESULT ON THE SIX TEST ROOMS: RGB ONLY, NO ALIGNMENT, MEASURED AGAINST FARO. THE TWO REFERENCE ROWS (EXP. 1) NEED A SENSOR AT RUN TIME.

Model	Teacher	Chamfer	Acc.	Compl.	F@5	R@10	AbsRel	within-1.25×	RMSE
DA-v2 Metric-Indoor, pretrained	— (Hypersim)	54.3 ± 10.6	70.3	38.3	0.03	0.10	0.382	0.277	49.3
+ fine-tune FT-LiDAR-24	iPad LiDAR, 24 rooms	15.7 ± 4.0	17.3	14.0	0.24	0.51	0.134	0.812	20.2
ref.: iPad LiDAR (sensor)	—	3.0 ± 0.6	2.2	3.9	0.93	0.97	0.018	0.997	3.9
ref.: DA-v2 L + 200 sparse pts/frame	—	6.0 ± 1.7	6.4	5.5	0.74	0.91	0.036	0.985	7.4

TABLE V

EXP. 6 PER TEST ROOM (ORDER OF TABLE II): CHAMFER (CM) AND ABSREL.

	4244	4733	4711	4289	4742	4526	mean
Depth Pro, Chamfer	131.1	165.2	172.9	196.4	177.9	88.9	155.4
pretrained, Chamfer	48.5	61.6	41.5	48.9	54.1	71.0	54.3
FT-Faro-10, Chamfer	10.8	13.8	17.6	21.1	12.4	11.9	14.6
FT-LiDAR-10, Chamfer	13.9	19.0	12.8	12.5	16.6	24.6	16.6
FT-LiDAR-24 , Chamfer	12.8	16.7	17.3	12.2	12.4	22.5	15.7
pretrained, AbsRel	0.375	0.346	0.255	0.412	0.476	0.430	0.382
FT-LiDAR-24 , AbsRel	0.127	0.156	0.193	0.087	0.141	0.101	0.134

TABLE VI

EXP. 7: 2D ONLY, 20 VALIDATION-FOLD ROOMS NEVER USED FOR ANYTHING ELSE (2,090 FRAMES); MEAN ± STD OVER ROOMS.

Row	AbsRel	within-1.25×	RMSE (cm)
DA-v2 Metric-Indoor, pretrained	0.410 ± 0.164	0.333 ± 0.216	50.9
+ fine-tune FT-LiDAR-24	0.130 ± 0.046	0.827 ± 0.135	18.4
ref.: iPad LiDAR (sensor)	0.019 ± 0.002	0.997 ± 0.002	3.8

C. Context: the price of each depth source (Exp. 1)

- 1) **The pipeline loses ~2cm; LiDAR adds ~1cm** (0.98 ± 0.25 cm, same in every room). The two differ clearly only at $\tau = 2$ cm: iPad LiDAR is wrong at the 2–4 cm level.
- 2) **Monocular depth with the right scale per frame (oracle) is 2.29 ± 1.11 cm behind LiDAR**, and **200 sparse metric points per frame recover it** (0.6 cm from the oracle, $3.8\times$ better than one-off calibration).
- 3) **One calibration per scene is $4.5\times$ worse than the oracle, uniformly** (19.9–26.0 cm in every room): a property of the model family, not of a failing room (Section VI-B).
- 4) **The pretrained metric model is worst** (54.3 cm): the surfaces are all there, in the wrong place. Kitchen 45261556 is among its best shapes under the oracle (4.5) and its worst metric room (71.0): right shape, wrong scale. This is the baseline FT-LiDAR-24 improves on.

D. Secondary findings (Exp. 2–5)

Table VIII summarises the rest. A $13\times$ smaller model (DA-v2 Small) costs only 1.0 cm for a $6\times$ speed-up, whereas one scale per scene costs 17.6 cm: model size is not the bottleneck, scale is. Frame count governs coverage, not accuracy, and LiDAR confidence maps give no measurable gain.

TABLE VII

EXP. 1: SAME PIPELINE, ONLY `DEPTH.SOURCE / DEPTH.ALIGNER` CHANGE (6 SCENES).

Source	Aligner	Chamfer	F@2	F@5	AbsRel
Faro (gt)	identity	2.1 ± 0.5	0.86	0.97	0
ARKit LiDAR	identity	3.0 ± 0.6	0.54	0.93	0.018
DA-v2 L	oracle_frame	5.3 ± 1.3	0.46	0.79	0.032
DA-v2 L	sparse_points	6.0 ± 1.7	0.38	0.74	0.036
DA-v2 L	per_scene	22.9 ± 2.1	0.08	0.23	0.237
DA-v2 Metric	identity	54.3 ± 10.6	0.01	0.03	0.382

TABLE VIII

SECONDARY EXPERIMENTS (CHAMFER IN CM, 6 SCENES).

Exp.	Result
2 voxel (gt)	2 / 4 / 8 cm: 1.2 / 2.1 / 2.8; 4 cm is used throughout (fusion 1–3 s per scan at every size)
3 frame stride	GT accuracy is constant (1.2–1.3 cm); fewer frames only lose coverage: F@5 cm 0.97 at 4 fps, 0.91 at 0.8 fps, 0.83 at 0.4 fps
4 model size (oracle)	DA-v2 L / S / MiDaS small: 5.33 / 6.35 / 13.24 at 871 / 121 / 27 ms per frame
5 LiDAR confidence	masking or weighting changes Chamfer by ≤ 0.03 cm; keeping only confidence 2 loses completeness

VI. DISCUSSION

A. Why fine-tuning works—and why it is not enough

We analysed every fifth frame of the six test rooms (471 frames): per frame, the GT/pred ratio and AbsRel under increasing amounts of scale help (Table IX). **Fine-tuning learns the domain’s multiplier.** The pretrained model is 15–31 % too far in every room; FT-LiDAR-24 is within ~20 % in both directions. FT-LiDAR-24 without calibration (0.134) is on par with the pretrained model given a per-room scale fitted on ground truth (0.127): what a one-off calibration provides has moved into the head’s weights, learned from LiDAR in other rooms. With the encoder frozen, shape barely changes (oracle 0.035 vs. 0.042). **The per-frame swing remains.** Within one room, frames at p5 and p95 still need factors $\sim 1.5\times$ apart, as before fine-tuning; with the right scale on every frame FT-LiDAR-24’s AbsRel would drop to 0.051. That information is not in a single image and does not come from a better teacher; it must come from metric data at run time. Fine-tuning and per-frame metric points are therefore complementary.

TABLE IX
PER-FRAME SCALE OF THE PRETRAINED METRIC MODEL VS.
FT-LiDAR-24 (SIX TEST ROOMS, EVERY FIFTH FRAME).

	Pretrained	FT-LiDAR-24
median(GT/pred) per room	0.69–0.85	0.93–1.21
per-frame swing within a room (p95/p5)	1.55×	1.52×
AbsRel, raw	0.381	0.134
AbsRel, one (S, T) per room from 10 GT frames	0.127	0.117
AbsRel, right scale on every frame	0.045	0.051
AbsRel, oracle (s, t) per frame	0.035	0.042

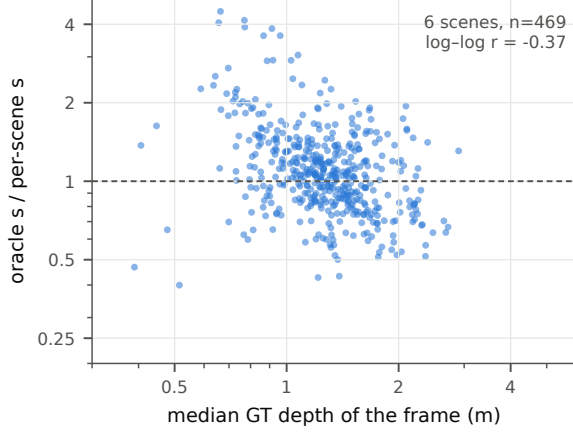


Fig. 4. Per-frame oracle scale divided by the per-scene scale (DA-v2 Large, all scenes) against the frame’s median GT depth: the scale a frame needs depends on how close the camera is to what it sees.

B. Scale follows the view, not time

Refitting (s, t) on every fifth frame of the relative model shows why one calibration per scene fails. Within one scan the central 90 % of frames need scales 2.6–4.9× apart; the correlation of scale with time flips sign between scenes (−0.33 to +0.30), so this is *not cumulative drift*, whereas the correlation with the frame’s median depth is negative in every scene (−0.27 to −0.75, Fig. 4). The worst frames are downward shots of table-tops at 0.5–0.8 m and frame-filling blank walls or cabinets: with no size cue the model guesses at “room scale,” and a table at 0.5 m lands at 1.5–2 m as a phantom surface. The metric model behaves the same way: right on average up to a per-room bias, wrong per frame (GT/pred from 0.43 to 1.35 within one room), but with almost no shift term—one scalar per frame recovers 97 % of its gap to the oracle. In 3D the same surface seen from several frames is placed at several distances and TSDF averages them into a thick slab.

Rooms that are hard for LiDAR are not those that are hard for monocular depth: the standard bedroom 47333774 is the easiest for LiDAR (2.4) but second-hardest for mono+oracle (6.2), because a small room scanned close to blank walls gives the model no cue. The mirrored bathroom is hard for every source whose scale is right (LiDAR 4.1, oracle 7.5), so a bathroom-renovation product should expect worse than the averages.

TABLE X
USE-CASE FIT: RECALL@ τ OVER 6 SCENES ($\checkmark \geq 0.9$, ~ 0.75 – 0.9 , $\times < 0.75$).

Use case (τ)	LiDAR	sparse	per_scene	FT-LiDAR-24	metric
overview (10 cm)	$\checkmark$ 0.97	$\checkmark$ 0.91	$\times$ 0.47	$\times$ 0.51	$\times$ 0.10
furniture (5 cm)	$\checkmark$ 0.91	$\sim$ 0.77	$\times$ 0.27	$\times$ 0.28	$\times$ 0.04
renovation (2 cm)	$\times$ 0.52	$\times$ 0.39	$\times$ 0.09	$\times$ 0.09	$\times$ 0.01

C. Implications for a product

Table X translates the results into product tiers. iPad LiDAR suffices for furniture placement but not for renovation measurement. On a device without LiDAR, neither one-off calibration nor a raw metric model serves any tier; FT-LiDAR-24 raises recall@10 cm from 0.10 to 0.51 with nothing at run time, a large step that is still short of the overview tier. With per-frame sparse points the same device reaches 6.0 cm, close to furniture placement. The order of investment the data suggest is therefore: **(1) per-frame scale from VIO points, (2) a fine-tuned model to remove the domain bias, (3) model size last**—investing in the model first returns <3 cm on a 22 cm base. Strategically, an app with Pro-device users can collect its own teacher during normal use (RGB, LiDAR depth and pose arrive together from ARKit), replacing a \$20k laser scanner; our data show no accuracy case for laser capture for this purpose. At 4 fps DA-v2 Large takes 3.6× the video length on a laptop and DA-v2 Small 0.5×; fusion is < 3 % of the time.

VII. LIMITATIONS AND FUTURE WORK

Reference and poses. The reference mesh is fused from Faro depth with the same poses as every row, so pose drift is excluded and a real phone system incurs additional error; completeness means complete relative to what Faro saw. **Scale of the evaluation.** Mesh numbers come from six rooms (one used in development) and 2D numbers from 26, all captured with one iPad model in homes; offices, empty rooms, low light and other phone cameras are absent. **Fine-tuning.** Checkpoints were selected on only three validation rooms, whose ranking (FT-LiDAR-24 > FT-LiDAR-10 > FT-Faro-10) does not reproduce on the test rooms; FT-Faro-10 and FT-LiDAR-10 differ in the number of views as well as the label; and only the head of one model was trained with one recipe. **Proxies and bounds.** per_scene calibrates on ground truth and sparse_points uses clean LiDAR samples instead of triangulated VIO features, so both are upper bounds. Depth Pro was run through one implementation on 640×480 input. Times are unoptimised PyTorch on Apple Silicon and are meaningful only as ratios.

Future work, in order of impact: (1) combine FT-LiDAR-24 with per-frame sparse points, which the 2D analysis bounds at AbsRel 0.051; (2) a controlled teacher comparison on identical frames and a larger validation set; (3) training data and evaluation on more phone cameras, including Android; (4) evaluation with real ARKit/ARCore feature points and a reference captured by our own app.

VIII. CONCLUSION

We asked whether fine-tuning a metric monocular depth model with a phone’s own LiDAR makes it better than its pretrained self, measured against a laser scanner, through a room-reconstruction pipeline in which the depth source is the single variable. **It does, in every room we can measure:** fine-tuning only the head of DA-v2 Metric-Indoor on 24 rooms (FT-LiDAR-24) lowers Chamfer from 54.3 cm to 15.7 cm on all six held-out rooms and per-frame AbsRel from 0.410 to 0.130 on 20 further rooms, with no calibration or sensor at run time. **The teacher need not be a laser scanner:** Faro and LiDAR teachers are not measurably different, and knowing the camera’s intrinsics is no substitute for in-domain labels. **It is not yet a sensor:** iPad LiDAR gives 3.0 cm and a monocular model scaled per frame on 200 sparse points 6.0 cm; FT-LiDAR-24’s remaining error is a scale that swings with what each frame shows, which fine-tuning de-biases but cannot remove. The path to a usable product on devices without LiDAR is therefore a fine-tuned model combined with per-frame metric points from the device’s tracker.

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